# The Two-Round Blockholder Disclosure Model

A standalone model note

v4 theory lane, ticket 30 (T2j)

Card stamp 2026-08-23, post-review repairs


**Abstract**

This note states the v4 two-round model in one place: primitives, timing, the partition the disclosure rule induces, the equilibrium notion, the standing hypotheses, and the eight-result ledger with its honesty labels. It then records what the v4 model takes from the frozen manuscript `draft_v2.tex`, what it simplifies, and what it drops; and it lists the executed evidence, failures included. Every statement of content here is transcribed from `research/model_v4/MODEL_CARD.md` (version stamp 2026-08-23, post-review repairs) and `research/model_v4/LABEL_LEDGER.md`. The card is the single source of truth: where a proof file and the card differ, the card wins. The conditionality attached to each result is not decoration. It is the result.


## 1 Position and object

The disclosure rule *is* the market's partition. The model asks how liquidity $\kappa$ moves bidder entry and the expected takeover premium when a stake threshold $\tau$ and a filing window $T$ split control-node histories into a **flagged** cell (the filing has landed before the control decision) and a **pooled** cell (it has not).

The control-outcome object is the expected engagement-related premium $\Delta^{\mathrm{act}}(\kappa, \tau, T)$. The price-path objects are the run-up path $R_d$, the cumulative run-up $R$, and the filing-day jump $J$. Lower $\tau$ means a tighter threshold margin; lower $T$ means a tighter window margin.

Throughout, $\kappa \in [0, 1]$ is *noise-trading intensity* — never depth, volume or turnover. Upright $T$ is the filing window; $\mathcal{T}$ is the outer best-response map. Bare $\lambda$ is not available (it is the appropriability coefficient of the tender-game appendix), and neither is a bare $C$, $W$, $\mathcal{S}$, $u$ or $g$: every one of those letters is overloaded in this project and always carries its subscript.

## 2 Primitives

### Fundamentals, signal, bidder

- $v \sim N(\mu_v, \sigma_v^2)$ is the target's standalone value; the blockholder observes $s = v + \varepsilon$ with $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ independent of $v$. The Gaussian projection coefficient is $\beta = \sigma_v^2/(\sigma_v^2 + \sigma_\varepsilon^2) \in (0, 1)$, so that $\mathbb{E}[v \mid s] = \mu_v + \beta(s - \mu_v)$. (The letter is $\beta$, the frozen manuscript's name; bare $\lambda$ is reserved.)
- The bidder draws a private synergy shock $\xi \sim N(0, \sigma_\xi^2)$ independent of $(v, s)$; $\bar{S}$ is mean bidder synergy and $K > 0$ the entry cost.
- $m_0, m_1$ are the takeover premia without and with engagement, with $m_1 > m_0$ *and* $m_0 \geq 0$; $\Delta_m = m_1 - m_0 > 0$ is the premium wedge, and $\Delta_V \geq 0$ is the non-takeover value engagement creates. The restriction $m_0 \geq 0$ is load-bearing: it is what makes the inner pricing fixed point exist, be unique and be continuous (Section 6, A5).

### Trading, stakes, the legal clock

- Noise is a ternary mark of size $\bar{z} > 0$: $\mathbb{P}(z_d = 0) = 1 - \kappa$ and $\mathbb{P}(z_d = \pm\bar{z}) = \kappa/2$.
- $b_0$ and $\bar{b}$ are the initial and maximum stake, $0 \le b_0 \le \bar{b}$, with $b_0 < \tau$ maintained: a pre-existing crossing is outside the core.
- $\mathcal{J}$ is a finite ordered menu of complete contingent plans, least to most aggressive, $|\mathcal{J}| = J < \infty$; $a_j \in \{0, 1\}$ is the engagement attached to plan $j$, with $a_j = 1$ for Voice plans and 0 for Exit and Hold.
- $B_j(s, d) \in [0, \bar{b}]$ is the cumulative pooled stake at business day $d$, with $B_j(s, -1) = b_0$; for Voice plans $\partial_d B_j \ge 0$ and $\partial_s B_j \ge 0$, Hold is constant and Exit weakly decreasing. For *every* plan and every $d$, $s \mapsto B_j(s, d)$ is Borel — automatic for Voice and Hold, a genuine requirement for Exit, and needed because pooled pricing integrates over every type including Exit types.
- $b_j^*(s) = B_j(s, H)$ is the terminal target stake; $H$ is the finite control-decision horizon in business days.
- $c_j(s; \tau) = \inf\{d : B_j(s, d) \ge \tau\}$ is the threshold-crossing date ($+\infty$ if it never happens); $f_j = c_j + T$ is the legal filing date; the flag lands iff $f_j \le H$, equivalently iff $B_j(s, H - T) \ge \tau$.
- $D_j(s; \tau, T) = \mathbf{1}\{a_j = 1,\ c_j < \infty,\ f_j \le H\}$; $D = 1 \Rightarrow a = 1$. $B_j^F = B_j(s, f_j)$ is the stake at filing and $Q_j^F = b_j^*(s) - B_j^F \ge 0$ the flagged-round order for a Voice plan.
- $\Gamma$ is a finite ordered coarsening from stake increment to pooled order mark, $q_{jd}(s) = \Gamma\big(B_j(s, d) - B_j(s, d - 1)\big)$, and observed pooled order flow is $X_d = q_{jd} + z_d$. ($\Gamma$, not $\psi$: $\psi$ is tender-game pivotality and $\chi$ is the frozen manuscript's cost parameter.)

### Information, prices, control outcome

$\mathcal{H}_d^P = (X_0, \ldots, X_d; \text{flag landed by } d)$ is the pooled public history, finite. $F = (B^F, a = 1)$ is the truthful filing message and $\mathsf{S}_F = (B^F, Q^F, a{=}1)$ is $F$ augmented by the flagged-round order. The control-node information set is

$$\mathcal{I}_H = \mathcal{H}_H^P \ \text{ on the pooled cell } \{D = 0\}, \qquad \mathcal{I}_H = \mathsf{S}_F \ \text{ on the flagged cell } \{D = 1\},$$

the bidder's own $\xi$ being private. The engagement posterior is $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I})$, equal to 1 on the flagged cell; entry probability is $p(\mathcal{I}) = 1 - \Phi\big((P + K + m_0 + \pi\Delta_m - \bar{S})/\sigma_\xi\big) \in (0, 1)$; and with entry indicator $\mathsf{B}$ the terminal shareholder payoff is

$$Y = (1 - \mathsf{B})(v + a\Delta_V) + \mathsf{B}\,(P + m_0 + a\Delta_m).$$

Prices are competitive, $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$, with the convention $P_{-1}^P := \mathbb{E}[Y]$ for the pre-trading pooled price (needed whenever $c = 0$, which $T = H$ forces on every flagged history). The genuine fixed point sits at control nodes: at an earlier pooled date $d < H$ the price is a tower expectation of already-solved control-node values, with no self-reference. $P_{\mathrm{ND}}(\mathcal{H}_{f^-}^P) = P_{f^-}^P$ is the *not-yet-disclosed* price at the same realised order flow — not a never-disclosed counterfactual, under which reading the run-up/jump identity acquires a residual term. Then

$$R_d = P_d^P - P_{c^-}^P, \qquad R = P_{f^-}^P - P_{c^-}^P, \qquad J = P^F - P_{\mathrm{ND}}, \qquad P^F - P_{c^-}^P = R + J \ \text{ exactly.}$$

$J$ is *not* claimed $\kappa$-invariant.

### The blockholder's objective

$$U_j(s) = \mathbb{E}\big[\, b_j^*(s)\, Y - \mathcal{C}_j^{\text{trade}} - a_j C_j(s) \ \big|\ s, j \,\big],$$

the expected terminal value of the position the plan builds, net of what it costs to build and to engage: $\mathcal{C}_j^{\text{trade}}$ is the execution outlay (increments valued at the pooled prices $P_d^P$ up to the plan's last pooled date, plus $Q_j^F(s)P^F$ when $D_j = 1$) and $C_j(s) \geq 0$ the engagement cost. Only two properties of $U_j$ are ever used: *plan-locality* — $U_j$ depends on $j$ only through the executed stake path, the prices paid on it, the terminal stake, the engagement flag and the cost — and *integrability*, $\mathbb{E}[\max_j |U_j|] < \infty$.

### Premium and sensitivity objects

$$h(\mathcal{I}) = \pi(\mathcal{I})\,p(\mathcal{I}), \quad h \geq 0,\ h(0) = 0; \qquad \Delta^{\text{act}} = \Delta_m\,\mathbb{E}[h(\mathcal{I}_H)] \ \geq 0;$$
$$M_F = \Delta_m \mathbb{E}[h \mid D = 1], \quad M_P = \Delta_m \mathbb{E}[h \mid D = 0]; \qquad \Omega = \mathbb{P}(D = 1) = \mathbb{P}(a = 1)\,\omega_a;$$
$$\mathcal{S} = |\partial_\kappa \Delta^{\text{act}}|, \quad \mathcal{S}_P = |\partial_\kappa M_P|; \qquad C_h(\bar{\pi}) = h(0) - 2h(\bar{\pi}/2) + h(\bar{\pi}).$$

$\omega_a = \mathbb{P}(D = 1 \mid a = 1)$ is the disclosed share of engagements, the calibration target; $\Omega$ is the unconditional flagged weight and is the frozen manuscript's $\omega_P$. $\bar{\pi}$ is the **upper support point** of the pooled engagement posterior in the $\mathrm{A}(\tau)$ representation — *not* the pooled engagement share. The share is the mean $\mathbb{E}[\Pi_\kappa]$, which under $\mathrm{A}(\tau)$ is $\kappa$-invariant (a mean-preserving spread) and is strictly below $\bar{\pi}$ in any non-degenerate case, equalling $\bar{\pi}/2$ only under level symmetry $A_0 = A_1$. Reading $\bar{\pi}$ as the mean forces a point mass at $\bar{\pi}$ with $A'_\kappa = 0$ and zero interior motion — degenerate, and excluded. $A'_\kappa$ is the common derivative of the $\mathrm{A}(\tau)$ weights ($A'_0 = A'_1 = A'_\kappa$, $A'_{1/2} = -2A'_\kappa$). Weight-effect ratios are $W_\tau, W_T$ (e.g. $W_T = (1 - \Omega(\tau, 5))/(1 - \Omega(\tau, 10))$) and composition-effect ratios are $C_\tau, C_T$ (e.g. $C_T = \mathcal{S}_P(\tau, 5)/\mathcal{S}_P(\tau, 10)$), unsigned.

For the general-equilibrium dominance-and-contraction node: $k = (k_1 \leq \cdots \leq k_{J-1})$ is the cutoff vector on the compact ordered polytope $\Theta$ with parameter vector $\vartheta$; $\mathcal{T}(k; \vartheta)$ the outer best-response map; $L_{\mathcal{R}} = \sup_{\mathcal{R}} \|D_k \mathcal{T}\|$ the contraction bound; $r_\tau = -\tau$, $r_T = -T$ the strictness coordinates (higher $r$ = tighter); $g_r^{PE} = -\operatorname{sgn}(d\Delta^{\text{act}}/d\kappa)\,\partial_{\kappa r}\Delta^{\text{act}}$ the direct fixed-policy attenuation margin; $\bar{k}_x = |\partial_x \mathcal{T}|/(1 - L_{\mathcal{R}})$ and $\bar{k}_{\kappa r}$ the inversion-free derivative bounds;

$$\mathcal{B}_r^{GE} = |\Delta_{\kappa k}|\bar{k}_r + \big(|\Delta_{kr}| + |\Delta_{kk}|\bar{k}_r\big)\bar{k}_\kappa + |\Delta_k|\bar{k}_{\kappa r}, \qquad \eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$$

the GE remainder bound and the slack on a dominance-and-contraction region $\mathcal{R}_r$.

# 3 Timing: the two rounds

1. Nature draws $v$ and the blockholder's signal $s$; the blockholder picks one complete contingent plan $j$ from the finite ordered menu $\mathcal{J}$.
2. **Round 1 — pooled trading.** The plan's stake path executes over business days $d = 0, \ldots, H$. Market makers see pooled order flow and set $P_d^P = \mathbb{E}[Y \mid \mathcal{H}_d^P]$. There is **no within-window re-optimisation**, and hence no feedback from realised order flow or prices into the path: $B_j(s, d)$, $q_{jd}(s)$ and $Q_j^F$ are functions of $(j, s, d)$ and $(j, s, \tau, T)$ alone. This is a numbered hypothesis, not a modelling aside: two steps of L2 fail without it.
3. **Disclosure node.** The flag lands iff $D = 1$, i.e. iff the plan engages, crosses $\tau$ at some date $c < \infty$, and $c + T \leq H$. The filing reveals $F = (B^F, a = 1)$. **The flag terminates the pooled round.** Pooled trading stops when the filing lands; the flagged round follows it and the bidder acts after that. Without this reading $Q^F = b_j^*(s) - B_j^F(s)$ is not the blockholder's whole residual position, and the flagged-round step of P1 fails.
4. **Round 2 — flagged trading, then the bidder.** If $D = 1$ the blockholder submits $Q^F$, the market prices $P^F = P(F, Q^F)$, and then the bidder decides. If $D = 0$ there is no flagged round and the bidder acts on the pooled history.

Sequence: pooled round → flag or no flag → flagged round if applicable → bidder decision.

# 4 The partition

The disclosure indicator $D = \mathbf{1}\{a = 1,\ c(\tau) + T \le H\}$ maps every control-node history into exactly one of two cells, $\mathcal{C}_F$ (flagged) and $\mathcal{C}_P$ (pooled), exclusive and exhaustive by construction. The cells are keyed by the disclosure rule and by nothing else:

- On $\mathcal{C}_F$ the control-node information set is the flagged tuple $\mathsf{S}_F = (B^F, Q^F, a{=}1)$: the market knows engagement happened, $\pi = 1$, and the pooled residual is (under A7′) pure noise conditional on $\mathsf{S}_F$.
- On $\mathcal{C}_P$ the control-node information set is the pooled public history $\mathcal{H}_H^P$: the market has an inference problem, and the posterior $\pi(\mathcal{H}_H^P)$ is where $\kappa$ enters.
- The clock equivalence: for every Voice plan, $f_j \le H \iff B_j(s, H - T) \ge \tau$. A tighter threshold (lower $\tau$) or a tighter window (lower $T$) moves histories from $\mathcal{C}_P$ into $\mathcal{C}_F$ and nowhere else.
- Each flagged history yields $B^F$, $R_d$, $R$, $J$, with $P^F - P^P_{c^-} = R + J$ exactly.
- $\Omega = \mathbb{P}(D = 1)$ is the cell weight; A8 asks for $0 < \Omega < 1$, which is required only for positive cell mass, never for the structural partition itself.

# 5 Equilibrium notion

**Cutoff perfect Bayesian equilibrium**: (i) a weakly ordered cutoff vector $k = (k_1 \le \cdots \le k_{J-1})$ mapping $s$ into a plan; (ii) sequentially optimal pooled and flagged components; (iii) Bayes-consistent beliefs on path; (iv) competitive pooled and flagged prices at their fixed points; (v) the bidder-entry rule; (vi) off-path beliefs as limits of full-support perturbations. Weak inequalities permit collapsed action regions, Hold included. Existence is Brouwer on the compact ordered polytope $\Theta$ for the outer map $\mathcal{T}(k; \vartheta)$, i.e. $k = \mathcal{T}(k; \vartheta)$. **Uniqueness is not claimed.**

# 6 Standing hypotheses

**A1 Independent primitives.** $v, \varepsilon, \xi$ and all $z_d$ mutually independent; all variances strictly positive.

**A2′ Finite model, amended boundedness.** Finiteness is unchanged: the plan menu $\mathcal{J}$, the image of $\Gamma$, the noise support $\{-\bar z, 0, +\bar z\}$ and the horizon $H$ are all finite. The old flat boundedness clause was *false* and is replaced by: prices and payoffs are locally bounded in $(s, \vartheta)$ on the maintained parameter set, and $\mathbb{E}[\max_{j\in\mathcal{J}} |U_j|] < \infty$ for every $k \in \Theta$. Flat global boundedness is inconsistent with the rest of the model — $v$ is Gaussian and the flagged region is unbounded in $s$ under A7′, so $Y$, prices and $U_j$ are unbounded. Integrability is all any proof consumes.

**A3 Ordered plans, single crossing.** At every belief/price system, adjacent-plan payoff differences cross zero at most once in $s$, and the preferred plan is weakly increasing in $s$.

**A4 Legal-clock discipline.** $c$ is the first date the path reaches $\tau$; the filing lands exactly at $c + T$; filings truthfully reveal stake and purpose; only Voice plans cross in the core.

**A5 Inner pricing regularity, mostly demoted to a theorem.** Each public-history pricing map has a unique fixed point, continuous in beliefs, cutoffs and parameters. *Under $m_0 \ge 0$ — a card restriction — existence, uniqueness and continuity of the inner fixed point are theorems, not assumptions*: with $\bar y(\mathcal{I}) = \mathbb{E}_\mu[v] + \pi\Delta_V$ and $\bar m(\mathcal{I}) = m_0 + \pi\Delta_m$, the pricing map is $P \mapsto \bar y + \bar m\, p(P)/(1 - p(P))$, strictly decreasing in $P$ wherever $\bar m \ge 0$, so it crosses the identity exactly once. A5 is retained only as its continuity clause: the pricing family is continuous in the cutoff vector and the parameters and measurable in the flagged tuple. Because the flagged information sets are continuum-indexed, "unique fixed point" must be read as a measurably selected *family*, not a finite list.

**A6 Compact outer self-map.** All best-response cutoffs lie in a common compact ordered polytope $\Theta$; $\mathcal{T}$ is continuous and maps $\Theta$ into itself.

**A7 / A7′ / A7-J Filing sufficiency.** On flagged histories $(B^F, Q^F, a = 1)$ identifies the informed component of the selected plan; conditional on it the pooled order-flow residual is pure noise, independent of $(v, s, \xi)$. L2 uses A7′ on path; the weak wording is not sufficient — it permits two $(j, s)$ pairs with different pooled paths, which is exactly L2's first failure case. The two injective forms are distinct. **A7′ (on-path composed target)** requires the composed terminal target $s \mapsto b^*_{j(s)}(s)$ to be strictly increasing on the flagged signal region, quantified over every cutoff vector $k \in \Theta$; strictness is required only for flag-capable composed targets, with no backtracking across admissible Voice-plan switches. **A7-J (joint tuple injectivity)** requires the full map $(j, s) \mapsto (B^F_j, Q^F_j, a_j)$ to be injective on the full flagged-pair set, including pairs not selected on path. A7-J is stronger than on-path A7′ and is the form consumed by the pre-review P1 proof. Strictness of $B^F$ is neither necessary (it fails at crossing-date jumps on the pro-rata menu) nor sufficient (multi-Voice backtracking). Under A7′ the flagged tuple is continuum-valued as a tuple: injectivity forces $(B^F, Q^F)$ to be continuum-valued, while the coordinates may trade the burden. Injectivity plus measurability already gives the measurable inverse (standard Borel spaces); no separate assumption is needed. *Satisfiability is resolved for A7′*: A7′ plus a fixed cutoff policy plus $\Omega > 0$ deliver the on-path injective form with an explicit inverse, and the pro-rata single-Voice menu with terminal target strictly increasing on all of $\mathbb{R}$ also satisfies A7-J. A7-J additionally needs strict increase off the Voice region; a target flat below the Voice cutoff breaks A7-J while leaving A7′ intact. Failure boundary: a binding stake cap, quantized stakes, a composed target repeating values across Voice-plan switches, $\Omega = 0$, and policy-dependence when the condition is stated at only one equilibrium's cutoffs. A7′-satisfying menus are fully separating on the flagged set, so the burden moves to incentive compatibility rather than away.

**A8 Interior crossing.** $0 < \Omega(\kappa, \tau, T) < 1$.

A($\tau$) **Threshold chord restriction.** The pooled posterior law has the symmetric ternary representation

$$\mathbb{E}[h] = A_0(\kappa)h(0) + A_{1/2}(\kappa)h(\bar{\pi}/2) + A_1(\kappa)h(\bar{\pi}), \qquad A_0' = A_1' = A_\kappa', \quad A_{1/2}' = -2A_\kappa',$$

with maintained orientation $C_h(\bar{\pi}) \le 0$ and $|C_h|$ weakly increasing in $\bar{\pi}$. Two clauses:

($\tau$-i) *The kernel depends on the information set only through the engagement posterior*: $h(\mathcal{I}) = h(\pi(\mathcal{I}))$, so $h(0)$, $h(\bar{\pi}/2)$, $h(\bar{\pi})$ are well defined and $\kappa$-free. This is a restriction, not a reading: $h = \pi p$ and $p$ depends on the price as well as on $\pi$, so in the model $h = \pi\, p(\hat{v}, \pi)$ is a function of *two* scalars. The clause says the standalone-value channel and the engagement channel do not co-move inside the pooled cell in a way that moves $h$ at a fixed posterior.

($\tau$-ii) *The support and $\bar{\pi}$ are $\kappa$-free; only the weights move.* Without the second half, L3's conclusion is **false**: the derivative gains a term first order in $\bar{\pi}$ and the vanishing fails.

*Where the bite is*: given a $\kappa$-invariant three-point support, the derivative restrictions are *equivalent* to $\kappa$-invariance of the pooled block's total mass and of its unnormalised engagement moment, both of which the model delivers at fixed policies. A($\tau$)'s entire remaining content is the support condition. A one-round ternary-noise market with informed mark $2\bar{z}$ and pre-order engagement share $\frac{1}{2}$ satisfies it; the frozen manuscript's own no-disclosure structure (informed mark $\bar{z}$) does **not** — its pooled law has four atoms, two of which move with $\kappa$. **Whether the two-round pooled cell of the timing above satisfies the support condition is OPEN.** Every A($\tau$)-conditional result — and therefore L4 leg 3 and T1 Part B — inherits that conditionality.

A(br) **Chord–sensitivity bridge.** Consumed by L4 leg 3 and T1 Part B, and by nothing else. For two compared thresholds $\tau' < \tau$ at fixed policies and a common $\kappa$:

(br-i) *Representation at both policies.* A($\tau$)'s symmetric ternary representation holds for the pooled class under $\tau$ *and* under $\tau'$, with endpoints $\bar{\pi}(\tau)$, $\bar{\pi}(\tau')$ and coefficients $A'_\kappa(\tau)$, $A'_\kappa(\tau')$.

(br-ii) *$\kappa$-localisation.* At fixed policies all $\kappa$-dependence of $M_P$ sits in the A($\tau$) weights, so $\partial_\kappa M_P = \Delta_m A'_\kappa C_h(\bar{\pi})$ exactly, with no composition-through-$\kappa$ remainder. Written against the honest reading $h = \pi\, p(\hat{v}, \pi)$, this repairs an ambiguity rather than adding a fourth independent restriction; the trailing "hence" is derivable, not assumed.

(br-iii) *Coefficient stability across the threshold margin.* $|A'_\kappa(\tau')| \leq |A'_\kappa(\tau)|$; weakest sufficient form is equality — reclassification changes *which* histories are pooled, not the $\kappa$-responsiveness of the pooled weights. *This is the clause with the least justification behind it; it is the one to attack first.*

(br-iv) *Endpoint linkage.* $\bar{\pi}$ is A($\tau$)'s chord endpoint — the upper support point — and is a weakly increasing function of the pooled prior engagement share $\bar{\pi}_{\text{pr}} = \mathbb{P}(a = 1 \mid D = 0)$, the same function at $\tau$ and $\tau'$. The identity branch $\bar{\pi} = \bar{\pi}_{\text{pr}}$ is excluded as degenerate.

(br-v) *Comparability of the chord functional across thresholds.* $C_h(\cdot)$ — and the kernel $h$ it is built from — are the same functions of the posterior at both compared thresholds. Without it, leg 3 compares $|C_h|$ across two different functionals and the comparison is meaningless. Required independently by three agents, and confirmed not implied by (br-i)–(br-iv).

*Sharpening on file, recorded not assumed*: $\bar{\pi} = \bar{\pi}_{\text{pr}}/\rho$ with $\rho := \frac{1}{2}A_{1/2} + A_1$ provably $\kappa$-free, so (br-iv) is equivalent to $\rho(\tau')/\rho(\tau) \geq \bar{\pi}_{\text{pr}}(\tau')/\bar{\pi}_{\text{pr}}(\tau)$. Under the level-symmetric reading $\rho = \frac{1}{2}$ and $\bar{\pi} = 2\bar{\pi}_{\text{pr}}$, forcing $\bar{\pi}_{\text{pr}} \leq 1/2$ — an inherited restriction on A($\tau$)'s domain that is not resolved.

**AGE GE differentiability and contraction.** On a candidate region $\mathcal{R}$ the outer map is twice continuously differentiable, $L_{\mathcal{R}} < 1$, and the sign of the equilibrium liquidity derivative is constant on $\mathcal{R}$.

# 7 The result ledger

Seven of eight results carry two-pass evidence — an adversarial proof-read PASS *and* an independent statements-only re-derivation PASS, by different agents — and are labelled PROVED with their conditionality written into the label. **P1 is CONJECTURE after the 2026-08-23 GPT end-review demotion; its earlier two-pass chain covered a mismatched A7 form.** The old evidence chain is retained, but it did not satisfy the two-pass gate for the recorded statement: the proof's h.7 consumes the joint injective form of A7, while the card row and the re-derivation carried the on-path form (form mismatch). The GPT end review and audit additionally identified a missing continuation-cost clause in Step 12 (sunk-cost gap, live for multi-Voice menus) and a false positivity claim at $\kappa = 1$. Ticket 35 owns the repair and any future re-promotion. Labels are transcribed from `LABEL_LEDGER.md`; each statement is the amended one, with its hypothesis set named in full. Nothing was weakened silently.

| ID | Statement, with its full hypothesis set | Label |
|---|---|---|
| D1 | Under **A1, A2′, A4, A5, the primitive/plan table restrictions, the cutoff selection map, and the pricing conventions** ($P^P_{-1} = \mathbb{E}[Y]$; $P_{\text{ND}} = P^P_{f^-}$), plus two hypotheses this regeneration wrote onto the card — **Borel regularity of** $s \mapsto B_j(s, d)$ **for every plan including Exit** (needed only for part (c), since pooled pricing integrates over all types) and **a content for** $\mathcal{I}_H$ — the indicator $D = \mathbf{1}\{a = 1,\ c(\tau) + T \le H\}$ is **measurable** and maps every control-node history into exactly one cell; for every Voice plan $f_j \le H \iff B_j(s, H - T) \ge \tau$; and each flagged history yields $B^F, R_d, R, J$ with $P^F - P^P_{c^-} = R + J$. | **PROVED** |
| L1 | Under **D1, the premium definitions, A5** (which pins *the* version of $\mathbb{E}[Y \mid \mathcal{I}]$), **A2′ with** $\Delta_m$ **finite, and A1** (one probability space): whenever $0 < \Omega < 1$, $\Delta^{\text{act}} = \Omega M_F + (1 - \Omega) M_P$; at $\Omega = 1$ it degenerates to $\Delta^{\text{act}} = M_F$ and at $\Omega = 0$ to $\Delta^{\text{act}} = M_P$, the null-cell average being *undefined rather than imputed* — proved as a non-identification statement, not asserted. | **PROVED** |
| L2 | At fixed cutoff and execution policies, under **A1; A2′ with the primitive/plan table restrictions; A4; A5; A7′ in its on-path injective form, consumed almost surely on the flagged set; D1; the no-feedback timing; and** $\Omega > 0$ — together with an explicit bidder-entry rule carried as bookkeeping: $(B^F, Q^F, a{=}1)$ makes the pre-filing pooled history conditionally independent of $(v, s, \xi)$ on the flagged set, so the flagged posterior, price, entry probability and $M_F$ are invariant to $\kappa$. ("Almost surely" is the only coherent reading once $B^F$ is continuum-valued, since then no individual flagged tuple has positive probability.) | **PROVED** |
| L3 | **Under** A($\tau$) — including ($\tau$-i) kernel-through-posterior and ($\tau$-ii) $\kappa$-free support *and* $\kappa$-free $\bar{\pi}$ — plus $h(0) = 0$; $\kappa$-free pooled mass and engagement moment at fixed policies; D1 by statement; minimal regularity ($h$ continuous on $[0, \bar{\pi}]$, twice differentiable on the open interval — Darboux does the rest, no continuity of $h''$); for the small-$\bar{\pi}$ corollary only, a second-order Peano expansion of $h$ at $0+$ and one and the same kernel along the shrinking family; and, for the seam where L4 consumes L3, $\lvert A'_\kappa \rvert$ bounded *uniformly in* $\bar{\pi}$. Then $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa C_h(\bar{\pi})$ exactly; $C_h(\bar{\pi}) = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$ for some $\zeta \in (0, \bar{\pi})$ — an identity, not an approximation; and $C_h = \frac{1}{4} h''(0)\bar{\pi}^2 + o(\bar{\pi}^2)$, so the interior motion vanishes at rate $\bar{\pi}^2$. **An "if", never an "iff"** ($A'_\kappa = 0$ also kills the motion). Whether the two-round pooled cell satisfies A($\tau$)'s support condition is **OPEN**. | **PROVED** under A($\tau$) |
| L4 | At fixed policies, for $b_0 < \tau' < \tau$ at a common window $T$ and common $\kappa$, with $\Omega(\tau', T) < 1$: **(leg 1, unconditional)** $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$ with every newly flagged history generated by a Voice plan, hence $\Omega(\tau', T) \ge \Omega(\tau, T)$; **(leg 2, unconditional)** the pooled engagement *share* falls, $\bar{\pi}_{\text{pr}}(\tau') \le \bar{\pi}_{\text{pr}}(\tau)$, with an exact identity for the gap; **(leg 3, under** A(br)**)** $\mathcal{S}_P(\tau', T) \le \mathcal{S}_P(\tau, T)$, with equality whenever $C_h(\bar{\pi}(\tau)) = 0$. Legs 1–2 need only **D1's clock equivalence, the no-feedback timing, fixed policies,** $b_0 < \tau' < \tau$ **imposed at both thresholds, A1, A4,** $D = 1 \Rightarrow a = 1$**, and** $\Omega(\tau') < 1$ — the "nestedness" clauses are *conclusions, not hypotheses.* Leg 3 additionally needs **L3 by statement,** A($\tau$)**'s maintained magnitude monotonicity of** $\lvert C_h \rvert$ (the sign half $C_h \le 0$ is never used at this leg) **and** A(br) **(br-i)–(br-v)**. | **PROVED** (legs 1–2 outright; leg 3 **under** A(br)) |

| ID | Statement, with its full hypothesis set | Label |
|---|---|---|
| P1 | Under **A1, A2′, A3, A4, A6, A7′ (on-path injective), D1 by statement, the no-feedback timing read with the flag-terminates-the-pooled-round clause, the definitional round-2 action-set hypothesis** (the flagged-round action set *is* the plan-generated set $\{Q^F_{j'}(s)\}$ over menu elements agreeing with $j$ on everything already played — *not* a closure condition, which is jointly unsatisfiable with finiteness by cardinality), $m_0 \geq 0$**, and the blockholder-objective definition** $U_j$: a cutoff PBE over complete contingent plans exists — $k^\star \in \Theta$ with $k^\star = \mathcal{T}(k^\star; \vartheta)$, prices at their inner fixed points, Bayes-consistent on-path beliefs, off-path beliefs as limits of full-support perturbations over *plans*, the entry rule, and a sequentially optimal flagged component. **A5 is not assumed**: its existence and uniqueness content is derived from $m_0 \geq 0$. **At any such equilibrium at which A8 holds**, both cells carry strictly positive probability and are on path; for A8's restatement as a single signal threshold add **H-ord** (Voice stake monotonicity across plans) and the upper-set engagement-flag hypothesis. A8 is a condition *on the fixed point exhibited*: existence of an equilibrium *at which* A8 holds is not claimed. Uniqueness is not claimed. | **CONJECTURE** |
| T1 | At fixed plan and cutoff policies, with $0 < \Omega < 1$ and $\mathcal{S}_P > 0$: **(A)** $\mathcal{S} = (1-\Omega)\mathcal{S}_P$ exactly, and the same factorisation holds for the total-variation aggregate of $\Delta^{\mathrm{act}}$ over any $\kappa$-grid with no differentiability required; **(B)** threshold tightening attenuates — $\mathcal{S}(\tau')/\mathcal{S}(\tau) = W_\tau C_\tau \leq 1$ because *both* ratios lie in $[0, 1]$, no dominance condition needed; **(C)** window tightening attenuates **iff** $W_T C_T \leq 1$, where $W_T \leq 1$ is *proved* (from D1's clock equivalence and the monotone Voice stake path) and $C_T$ is *unsigned* — "equivalently $\partial_{r_T}\mathcal{S}_P/\mathcal{S}_P \leq \Omega_{r_T}/(1-\Omega)$" holds **on average along the tightening path** (integrated over $[-T, -T']$), exactly in the infinitesimal limit, and is **false read pointwise**. Hypotheses: **fixed policies; A8 at each compared policy;** $\mathcal{S}_P > 0$**; L1; L2 (its own hypotheses travelling); D1; PE-**$\Omega$ **(**$\partial_\kappa \Omega = 0$ **at fixed policies — derivable, not assumed, and it fails in GE, which is C1's term);** $\kappa$**-differentiability of** $M_P$ **(no card hypothesis supplies this — carried in-proof);** A$(\tau)$ **at both compared policies with the** $\bar{\pi}$ **ruling; L3;** A(br) **(br-i)–(br-v) at the threshold pair; L4; the no-feedback timing; a smooth window interpolation for the local form; threshold-side smoothness (confirmed non-load-bearing)**. **No unconditional window sign is claimed.** | **PROVED** at fixed policies |
| C1 | On a named region where $L_{\mathcal{R}} < 1$ and $g_r^{PE} > \mathcal{B}_r^{GE}$, the fixed-policy attenuation sign survives in equilibrium. The dominance-and-contraction implication is **two-pass complete** — an adversarial proof-read PASS (0 FAIL, 13 repairs, 7 observations) and an independent statements-only re-derivation PROVED-WITH-CHANGES (a norm convention and two-sided openness of $\mathcal{R}_r$ added; at $J - 1 \geq 2$ the bare $\lvert \cdot \rvert$ in $\mathcal{B}_r^{GE}$ is not well defined and a mismatched reading makes part (B) false) — and the executed 80-node run returns **18 pointwise dominance-and-contraction nodes**, slack $\eta_r$ from 0.0595 (min) through 0.3467 (median) to 1.7227 (max), $L_{\mathcal{R}} \leq 0.5008$ at every node. These nodes verify the pointwise inequalities and supporting diagnostics only; they do not verify the full C1 antecedent or a named nonempty region. | PROVED |

Remaining aspiration, not a claim: promotion from pointwise dominance-and-contraction nodes to a named nonempty region, which would require the D8 $\varepsilon$-ball and integral-control pattern.

### Labels

**PROVED** — a complete proof, independently re-derived and proof-read. **NUMERICAL** — verified on a grid by an executed, committed check script with committed output. **ESTIMATED** — an empirical estimate with a standard error and a stated design. **CONJECTURE** — everything else, including anything whose proof is deferred. A **dominance-and-contraction node** is not a fifth label: it records pointwise $L_{\mathcal{R}} < 1$ and $\eta_r > 0$ with supporting diagnostics, not verification of the full C1 antecedent. A named-region promotion is not claimed. Labels are never weakened by editing; only an executed check or an independent re-derivation may move one.

## 7.1 Three known facts on file

These are facts the record carries, not claims the model makes. They are the reason the labels above are written with their conditions attached.

**1. O-1 is a disclosure-regime analogy, not a window-margin test.** This is a known fact on file, not a claim the model note makes: O-1 compares the public buy flagged versus pooled at fixed policies in the static repo model. Its ratios

$$1.06397\,/\,1.18373\,/\,1.13631\,/\,0.37798 \qquad \text{at} \qquad \Omega \;=\; 0.037252\,/\,0.128950\,/\,0.285804\,/\,0.50$$

are regime-comparison composition outcomes, not $W_T C_T$ and not a window pair. The analogy is useful because it shows that a composition factor can exceed one, motivating T1's genuine window-margin iff. The genuine window-margin record is `t2_t1_check` block 4: $W_T C_T < 1$ at every checked node at this calibration. The O-1 boundary $\Omega^\star \approx 0.3428$ remains a disclosure-regime boundary, not a window boundary. Provenance: this is the referee's O-1 experiment run on the *repo* model (the frozen manuscript's static structure, fixed cutoffs, partial equilibrium), re-executed in `quality_reports/fixes/t1_o1_rerun_check.py` with every committed number reproducing to the last printed digit; it is not the two-round build.

**2. Whether the implemented two-round pooled cell satisfies** $\mathrm{A}(\tau)$ **is OPEN at baseline.** Two executed diagnostics are informative but neither decides the support condition, and both are retained without smoothing.

- `t2_l2_check.json`, check `l2_placebo_M_P_sign_A_tau`: **FAIL** as a sign placebo. The enumerated pooled $M_P$ is *hump-shaped* in $\kappa$ — 10 of 18 increments positive, one sign change, peak near $\kappa = 0.55$. But $\mathrm{A}(\tau)$'s maintained $C_h(\bar{\pi}) \le 0$ does not sign $A'_\kappa$; the card's own Example A has $A'_\kappa = -1/4$. The demanded sign is therefore not implied by $\mathrm{A}(\tau)$, and the placebo is misformulated rather than a refutation of the support representation.
- `t2_t1_check.json`, check `t1_block3_chord_magnitude`: **FAIL** for the hard-coded Example-A coefficient. The residual $\big|\mathcal{S}_P - \Delta_m |A'_\kappa|\, |C_h(\bar{\pi})|\big|$ is 0.006279 (0.6279 premium percentage points) against a predicted $10^{-10}$ — the implied coefficient is $|A'_\kappa| \in [0.997, 1.158]$, while the check hard-coded 0.25. The ratio $|C_h|/\bar{\pi}^2$ remains constant to 0.48% between the two smallest $\bar{\pi}$ nodes, well inside the 5% criterion. This rejects the hard-coded calibration, not $\mathrm{A}(\tau)$'s general support representation.

The decisive support-enumeration check is ticket 33. Until it lands, $\mathrm{A}(\tau)$'s applicability at the implemented pooled cell remains OPEN; the executed diagnostics are test-design artifacts, not a wiring error or a theorem failure.

**3. L4's sign predictions hold numerically with zero violations.** `t2_l4_check.json` reports 10 checks, 0 failures. Across all 16 tightening steps and both windows: $\Omega$ rises at every step (0 violations, minimum increment $+0.02263$); $\bar{\pi}_{\mathrm{pr}}$ falls at every step (0 violations, largest increment $-0.018766$); and $\mathcal{S}_P$ falls at every step (0 violations, largest increment $-9.47 \times 10^{-5}$) — and, as the JSON notes, a single violation would be a failed hypothesis rather than sampling error, because nothing in that computation is sampled. $\Omega$ and $\bar{\pi}_{\mathrm{pr}}$ are flat in $\kappa$ to exactly zero across 71 grid points, which is what the no-feedback timing predicts.

**What the pair of facts means.** The chord mechanism is **hypothesis-bound**: it needs A($\tau$), whose applicability at the implemented pooled cell is OPEN. The threshold phenomenon is **not**: the reclassification legs of L4 are unconditional, and they hold numerically without exception.

# 8 What this note does not claim

A global window-margin attenuation sign; $\kappa$-invariance of $J$; equilibrium uniqueness; a named nonempty GE region *as a theorem* (the 18 pointwise dominance-and-contraction nodes are an executed record, not verification of the full C1 antecedent); endogenous filing before the deadline; noisy or partially revealing flagged-round trading; continuous-time execution; welfare or optimal rule design; that the frozen manuscript's hump result survives; that the prior calibration ($\Omega \approx 0.037$) is economically meaningful; any empirical value for $\omega_a$.

Three items are explicitly open:

1. **Whether the two-round pooled cell satisfies** A($\tau$) **— OPEN.** L3 proves the representation's entire remaining bite is the support condition, exhibits a one-round market that satisfies it and the frozen manuscript's own no-disclosure structure that does not, and names the weakest sufficient conditions. L3, L4 leg 3 and T1 Part B are all conditional on A($\tau$), so this is the largest single conditionality the ledger carries — and the executed diagnostics of fact 2 above do not decide it at baseline; the support-enumeration check is ticket 33.
2. **Whether an equilibrium in which the blockholder chooses the fully separating plan exists on a given calibration — OPEN, and P1-adjacent.** A7$'$ menus are fully separating on the flagged set, so the burden did not disappear when A7 satisfiability was resolved; it *moved* to incentive compatibility, which P1 does not settle.
3. **O-1 is a disclosure-regime analogy, not a window-margin test** (fact 1 above). The genuine window-margin record is `t2_t1_check` block 4, with $W_T C_T < 1$ at every checked node at this calibration.

# 9 What v4 takes from `draft_v2`, what it simplifies, what it drops

The frozen manuscript `draft_v2.tex` is a record the supervisor has seen; it is not edited and it is not cross-referenced here. The table below describes its objects rather than citing them.

| `draft_v2` object | Verdict | What carries over, what changes |
|---|---|---|
| Four-action structure: Exit, Hold, Quiet Voice, Public Voice, with weakly ordered cutoffs $(k_1, k_0, k_D)$; and the dominance lemma making the four-action menu a result rather than an assumption (engaged exit and silent buy are never chosen). | **REUSE** | Becomes the *special case* of the v4 ordered plan menu: $\mathcal{J}$ finite and ordered with $J \geq 3$, cutoff vector $k = (k_1, \ldots, k_{J-1})$, which maps to $(k_1, k_0, k_D)$ exactly when the menu is the four named actions. What changes: menu elements are now *complete contingent plans* over the calendar $d = 0, \ldots, H$, not single one-shot $(q, a)$ pairs, and A3's single crossing is stated for adjacent plans. The dominance argument survives as the reason the menu is ordered and finite. |
| Brouwer existence architecture: boundedness of best responses, the self-map and order-preservation lemma, and existence of a monotone equilibrium by Brouwer on a nonempty compact convex $\Theta$. | **REUSE** | Reused whole as P1's architecture: $\Theta$ compact, ordered, convex; $\mathcal{T}$ continuous $\Theta \to \Theta$; $k^\star = \mathcal{T}(k^\star; \vartheta)$; collapse faces permitted (the manuscript's own baseline collapses Hold). What changes: the fixed point ranges over plans; off-path beliefs are limits of full-support perturbations *over plans*; and the inner pricing fixed point is no longer an assumption (see the A5 row). |
| The (C*) chord condition $\mathcal{C}(\bar{\pi}) = h(0) - 2h(\bar{\pi}/2) + h(\bar{\pi})$, and the lemma signing the posterior/pricing channel under it. | **REUSE** | Is v4's $C_h(\bar{\pi})$, same expression, same maintained orientation $C_h \leq 0$ with $\lvert C_h\rvert$ weakly increasing in $\bar{\pi}$. What changes: in `draft_v2` it selects hump versus trough in $\kappa$; in v4 it is the magnitude that drives pooled sensitivity through L3's exact identity $\partial_\kappa \mathbb{E}_\kappa[h] = A'_\kappa C_h(\bar{\pi})$, and the A($\tau$) representation that licenses that identity is stated as a restriction whose support condition is open. |
| The tender-game microfoundation of the premium wedge (the disagreement-node continuation deriving the appropriability coefficient $\lambda = 1 - q(1-\gamma)\psi$ and the resulting wedge). | **REUSE** | Kept as the *input* that supplies $(m_0, m_1)$. v4 takes $m_0, m_1$ as primitives with $m_1 > m_0$ and $m_0 \geq 0$ and does not re-derive them; the tender game is where those numbers come from and why $\Delta_m > 0$ is not free. Its symbols are reserved and unavailable: bare $\lambda$ is the appropriability coefficient, $\psi$ is pivotality. |
| The GE dominance-check pattern: contraction modulus $L$ along the path, an inversion-free Neumann bound $\bar{B}$, pointwise dominance off a ball plus a ball-integral condition, a certified interval, and a counterexample showing the global claim false. | **REUSE** | Is C1's precedent. $\mathcal{B}_r^{GE}$ is the *cross-derivative* analogue of $\bar{B}$; $\eta_r = g_r^{PE} - \mathcal{B}_r^{GE}$ is the slack; the executed record names pointwise dominance-and-contraction nodes, not a verified region, and a named-region promotion remains open. What changes: the object bounded is a cross-derivative in $(\kappa, r)$ rather than a single-argument profile, and the strictness coordinate $r_T = -T$ is not differentiable because the window is an integer. |

| `draft_v2` **object** | **Verdict** | **What carries over, what changes** |
|---|---|---|
| Stake-triggered disclosure as the market's partition, with the flag a function of the order ($D = \mathbf{1}\{q = +1\}$: a buy cannot be concealed). | **REUSE** | The idea that the rule *is* the partition is the whole v4 position. What changes is the trigger: $D = \mathbf{1}\{a = 1,\ c(\tau) + T \leq H\}$, a stake threshold plus a filing window on a business-day calendar, with D1's clock equivalence $f \leq H \iff B(s, H - T) \geq \tau$. The window $T$ becomes a genuine primitive instead of a flag that is either on or off. |
| Assumption (A2)'s flat boundedness of prices and payoffs. | **SIMPLIFY** | Replaced by A2′: finiteness clauses kept verbatim; boundedness weakened to local boundedness plus integrability $\mathbb{E}[\max_j \lvert U_j \rvert] < \infty$. The flat bound was *false* in this model ($v$ Gaussian, flagged region unbounded in $s$), and no proof ever used it. |
| Assumption (A5)'s inner pricing regularity (existence, uniqueness and continuity of the price fixed point, assumed, with a $\delta/\sigma_\xi < 1/\phi(0)$ style regularity condition). | **SIMPLIFY** | Demoted to a theorem under $m_0 \geq 0$: the pricing map is strictly decreasing in $P$ wherever $\bar{m} \geq 0$ and crosses the identity exactly once. A5 survives only as its continuity clause, read as a measurably selected family because the flagged information sets are continuum-indexed. Three independent confirmations, including executed counterexamples that produce zero roots and three roots once $m_0 < 0$. |
| The discount factor $\delta$ inside the pricing fixed point $P = \delta[(1-p)\hat{V} + p(P+\bar{m})]$, and the sensitivity analysis around it. | **DROP** | v4 prices are undiscounted conditional expectations $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$. The regularity work $\delta$ was doing is done instead by $m_0 \geq 0$. |
| The one-round pooled posterior: $\pi(X, 0)$ in closed form from a single order-flow draw $X \in \{-2, -1, 0, 1\}$ with $p_0 = 1 - \kappa$, $p_1 = \kappa/2$ and the action weights $(\omega_E, \omega_H, \omega_Q)$; Hold and Quiet Voice pooling because both trade $q = 0$. | **DROP** | Replaced by a pooled history $\mathcal{H}_H^P$ over $H + 1$ business days with plans, $\Gamma$-coarsened increments and a flag that can land mid-calendar. The closed-form posteriors do not survive the calendar. Consequential: L3 shows the manuscript's own no-disclosure structure (informed mark $\bar{z}$) yields a *four-atom* pooled law, two of whose atoms move with $\kappa$, so it lies outside A($\tau$) — the one-round posterior is not merely superseded, it is a named example of the representation failing. |
| The hump headline: $\Delta^{\min}(\kappa)$ single-peaked in liquidity, as the conditional-hump proposition and the paper's first main result. | **DROP** | Not claimed anywhere in v4; the card states outright that it does not claim the hump survives. The v4 headline is the partition and the attenuation factorisation $\mathcal{S} = (1 - \Omega)\mathcal{S}_P$, not a shape in $\kappa$. Note the hump has not been *refuted* either: the enumerated two-round $M_P$ is itself hump-shaped in $\kappa$ at baseline (peak near 0.55) — it is simply no longer the result being sold, and its role now is to break A($\tau$)'s orientation clause. |

| `draft_v2` object | Verdict | What carries over, what changes |
|---|---|---|
| The unconditional attenuation claim: $\lvert\partial\Delta^{\mathrm{act}}/\partial\kappa\rvert$ decreasing in $\omega_P$, asserted as a monotone consequence of shifting mass to the disclosed cell, with no composition term and no margin distinction. | **DROP** | Split and conditioned. The weight effect is kept and proved ($W_T \leq 1$, $W_\tau$ likewise), but the composition effect is a separate ratio: T1(B) gets threshold attenuation *unconditionally* because *both* $W_\tau$ and $C_\tau$ lie in $[0,1]$; T1(C) makes the window margin an **iff**, $W_T C_T \leq 1$, with $C_T$ unsigned. O-1 is a disclosure-regime analogy whose composition ratio can exceed one; it is not a window measurement. The genuine window comparison is `t2_t1_check` block 4. An unconditional window sign is not available. |
| Equilibrium uniqueness as a numerical regularity (a 30-seed multistart converging to one fixed point per $\kappa$, labelled a regularity rather than a theorem). | **DROP** | v4 does not claim uniqueness at all, as a theorem or as a regularity. The multistart survives only as an existence diagnostic in the check scripts, and it does not always clear its own binding criterion (Section 10, P1). |
| The welfare apparatus: transfer netting and the present-value form of total surplus, the liquidity planner's wedge $\kappa^*$ versus $\kappa^\dagger$, the disclosure-threshold planner, the first-best benchmark. | **DROP** | Out of scope by the card's own statement: no welfare, no optimal rule design. Note this is also why a bare $W$ is unavailable as a symbol — it was total surplus there, and it is the weight-effect ratio $W_\tau, W_T$ here. |

# 10 Evidence

## The check-script inventory

Eight scripts, each with committed JSON output, in `quality_reports/fixes/`. Verdicts are as the JSON reports them, failures included. "Wiring" and "substantive" are the design's own classification: a wiring check runs both sides of an identity through the same enumeration and is therefore *not* evidence for the result, only for the code.

| Script | Verdict | What it establishes, and what it does not |
|---|---|---|
| `t2_d1_check` | **PASS** 9 checks, 0 FAIL, 1 vacuous | Clock equivalence by three independent routes (date-by-date scan, one evaluation at $H-T$, closed-form crossing date) agreeing to exactly 0; partition exclusivity and exhaustion at 0 residual; the $T = H$ corner; robustness at $H = 12$. 50 nodes. *Vacuous*: the $Q^F$ window-monotonicity check, because $\max Q^F = 0$ at the implemented calibration — the flagged round carries no residual order there, so the prediction has nothing to bite on. |

| Script | Verdict | What it establishes, and what it does not |
|---|---|---|
| `t2_l1_check` | **PASS** 4 checks, 0 FAIL | Decomposition residual $\Delta^{\mathrm{act}} - (\Omega M_F + (1-\Omega)M_P)$ at 0, both degenerate endpoints, and the non-evaluability of the null-cell average. *All four are classified WIRING by the design*: both sides run through the same enumeration, so the residual is machine noise by construction and is **not evidence for L1**. |
| `t2_l2_check` | **FAIL** (1 of 5) | L2's own content passes exactly: the flagged range of $M_F$ over 19 $\kappa$ values is 0 and its $\kappa$-derivatives are 0 — the flagged path never touches the $\kappa$-dependent array. Two placebos confirm the pooled side does move. **The failure is the fifth check**, an *ancillary* A($\tau$)-orientation placebo: see Section 6, fact 2. Because A($\tau$) does not sign $A'_\kappa$ (Example A has $A'_\kappa = -1/4$), the placebo is misformulated and does not decide A($\tau$)'s support condition; it is not a failure of L2. |
| `t2_l3_check` | **PASS** 10 checks, 0 FAIL | The derivative identity, the mean-value form $C_h = \frac{1}{4}\bar{\pi}^2 h''(\zeta)$, the quadratic rate as $\bar{\pi} \downarrow 10^{-4}$, the affine-kernel zero-chord case, and three failure witnesses (a tent kernel with no root, the four-atom Example B gap, an unbounded-$h''$ witness). *Scope*: it runs on the three- and four-atom analytic laws and the standalone chord module — it does *not* enumerate the two-round model, so it verifies L3's mathematics, not A($\tau$)'s applicability. |
| `t2_l4_check` | **PASS** 10 checks, 0 FAIL | All three sign legs with **zero violations** over 16 tightening steps (Section 6, fact 3); the exact reclassification identity at $10^{-16}$; flatness in $\kappa$ at exactly 0 over 71 nodes; the $\tau$-grid spanning a ninefold range in $\omega_a$; L3's quadratic corollary within 2.1%. Prediction 5 (the size of the $A'_\kappa$ channel) is *reported, not gated*, by the request's own instruction: median absolute residual 0.0352, max 0.1213. |
| `t2_p1_check` | **FAIL** (1 of 10) | Nine substantive checks pass: inner-root single crossing and transversality, the flagged family single-valued with the right slope, sequential optimality of the flagged component, both cells on path, the threshold reformulation, monotonicity of $\Omega$ in $(\tau, T)$, and *existence at the core nodes* under the full 30-seed multistart. **The failure is the grid sweep**: 23 of 27 policy nodes met the binding payoff-scale criterion ($10^{-9}$) and 4 did not, at best payoff scales $1.5 \times 10^{-3}$, $1.1 \times 10^{-3}$, $4.0 \times 10^{-4}$, $3.1 \times 10^{-4}$ — all four at extreme liquidity $\kappa \in \{0.15, 0.85\}$, at $\tau \in \{0.05, 0.075\}$ and $T \in \{5, 1\}$ (the independent re-run corrected an earlier gloss here: these are *not* zero-engagement-share or $T{=}H$ corner nodes — every actual corner row converged, and the recorded pooled prior engagement share is 0.102). Seed coverage is *ruled out* as the cause: the check re-ran all four nodes at the full 30 seeds and found identical best residuals, with most seeds landing on the same cutoff vector. All 27 nodes do converge on the cutoff-scale criterion. The cause is undiagnosed — candidates are a failure of A3's single crossing at those $\kappa$ extremes (the adjacent-plan payoff gap is a sawtooth on this menu, and single crossing is a calibration fact, not structural) or a genuine boundary non-existence outside the repaired P1 hypothesis set; P1 is **CONJECTURE** after the form-mismatch, sunk-cost, and $\kappa = 1$ review findings, and the four nodes are named, not smoothed. |

| Script | Verdict | What it establishes, and what it does not |
| --- | --- | --- |
| `t2_t1_check` | **FAIL** (1 of 9) | The factorisation $\mathcal{S} = (1-\Omega)\mathcal{S}_P$ holds pointwise to $1.2 \times 10^{-16}$ and in total variation to $2.2 \times 10^{-19}$; $\Omega$ is flat in $\kappa$ at 0 residual; the threshold margin, the window margin, the O-1 benchmark and the composition factors all pass; $H = 12$ robustness passes. **The failure is block 3**, the chord-magnitude route with hard-coded Example-A coefficient 0.25: see Section 6, fact 2. It is not a refutation of A($\tau$)'s general support representation. One check is *vacuous* (the local form, which needs a smooth window interpolation the integer calendar does not provide). |
| `t2_c1_region_check` | **PASS** 12 checks, 0 FAIL, pointwise nodes *nonempty* | 80 nodes ($8\ \kappa \times 5\ \tau \times 2\ T$), 2.9 hours of wall time. Two gated verdicts, both PASS with 0 violations: the bound contains the four-corner re-solve remainder at the 8 validation nodes (max ratio 0.511), and it contains the implicit-function remainder at all 80 (max ratio 1.0). **18 pointwise dominance-and-contraction nodes**, all at $T = 5$ and the upper three $\tau$ percentiles; $\eta_r \in [0.0595, 1.7227]$, median 0.3467; $L_{\mathcal{R}} \in [0.264, 0.501]$ with no node at $L \geq 1$. These nodes verify the two inequalities and supporting diagnostics only, not the full C1 antecedent or a named region. *Failure attribution*, recorded not hidden: 56 nodes have $g_r^{PE} = 0$ to $10^{-10}$ because the implemented legal clock quantises the crossing date, so $\Omega(\tau)$ is locally constant and the cross-derivative vanishes identically there; 4 are positive but dominated; 2 are negative; 40 nodes (all $T = 10$) are degenerate on flagged mass below 0.01. The sharper, non-inversion-free bound would certify 22 rather than 18 — reported as a measure of the Neumann step's price, and certifying nothing. No tolerance was weakened to make the pointwise node set nonempty. |

**Provenance caveat.** Seven of the eight scripts record the card stamp *2026-08-20* in their provenance block; only `t2_c1_region_check` was run against the 2026-08-21 stamp. The card changes between those stamps are hypothesis-explicitness moves, not model changes, but the mismatch is on the record and is not papered over here.

### The `numerical_v4` package: smoke facts

The implementation is `numerical_v4/` (modules `params`, `menu`, `policy`, `flagged`, `pooled`, `premium`, `solver`, `smoke`), with committed output in `numerical_v4/smoke_output.txt`. At the smoke configuration $H = 10$, $T = 5$, $M = 2$ order marks, $J = 3$ plans, $b_0 = 0.03$, $\bar{b} = 0.1$, seed $\tau = 0.05$ (the statutory 13D 5%):

- **Enumeration gate**: 4,194,304 histories, 826,686 feasible, $N_\theta = 12$; feasible $\times\ N_\theta = 9.92 \times 10^6$ against a gate limit of $10^8$. PASS. One evaluation takes 0.372 s; the whole smoke runs in 64.8 s.
- **$\tau$ freezing**: the seed equilibrium is $k = (1.31246481,\ 1.41888743)$ with $|k - \mathcal{T}(k)| = 1.34 \times 10^{-11}$; the median Voice $b^*(s)$ is 0.09076406, and $\tau$ is frozen there at every node.
- **Baseline equilibrium** ($\kappa = 0.5$): $k = (1.2405757283,\ 1.5310222869)$; outer residual $2.60 \times 10^{-11}$ on the diagnostic cutoff scale and *exactly zero* on the binding payoff scale.
- **Masses**: $\Omega = 13.84\%$, $\mathbb{P}(a = 1) = 22.63\%$, $\omega_a = 61.15\%$, $\bar{\pi}_{\text{pr}} = 10.21\%$. *Note $\omega_a = 61\%$ here is a property of the calibration, not an estimate*: the card claims no empirical value for $\omega_a$, and the empirical anchor for it is absent from the literature.

- **Premium objects**: $M_F = 0.5528$ pp, $M_P = 0.2228$ pp, $\Delta^{\text{act}} = 0.2685$ pp; L1's decomposition residual is 0 (wiring).
- **Price path**: mean run-up $R = 389.18$ bp, mean filing jump $J = 2590.52$ bp, D1 identity residual 0 against a $10^{-12}$ tolerance.
- **A7 certificate**: passes — minimum $|db^*/ds|$ on the flagged set is $2.20 \times 10^{-4}$ (gate $10^{-8}$), no flat sub-intervals, no $(B^F, Q^F)$ collisions, no cross-plan collisions, 10 flagged atoms of 19.
- **Numerical hygiene**: pooled mass error $2.2 \times 10^{-16}$; maximum inner price residual $2.2 \times 10^{-16}$; maximum $b^*$ inversion error $5.9 \times 10^{-14}$; GL-20 versus GL-40 quadrature on $M_F$ differing by $2.6 \times 10^{-18}$; **zero multiple-root nodes** anywhere, which is the structural consequence of $m_0 \geq 0$.
- **L2 verdict, substantive**: the range of $M_F$ across the frozen-policy $\kappa$ sweep is *exactly* 0.
- **The shape that breaks** A($\tau$): $M_P$ is hump-shaped in $\kappa$ with its peak near $\kappa = 0.55$, so $\mathcal{S}_P = |\partial_\kappa M_P|$ is V-shaped about that peak, ranging from $4.99 \times 10^{-3}$ to $2.66 \times 10^{-1}$ premium pp per unit $\kappa$; at the peak $\mathcal{S} = (1 - \Omega)\mathcal{S}_P = 6.37 \times 10^{-2}$.
- **Chord module**: $C_h \leq 0$ at every $\bar{\pi}$ tested, and $|C_h|/\bar{\pi}^2$ settles at $\approx 0.2219$ as $\bar{\pi}$ falls from $10^{-1}$ to $10^{-4}$ — L3's quadratic rate, on the standalone route.
- **One structural caveat**: $\max Q^F = 0$ at this calibration. The flagged round exists in the timing and in the code, but carries no residual order here, which is why the $Q^F$ monotonicity check is vacuous.

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Sources, all in the `v4-theory` worktree: `research/model_v4/MODEL_CARD.md` (stamp 2026-08-23, post-review repairs) for Sections 1–6 and 7; `research/model_v4/LABEL_LEDGER.md` for the labels; `research/model_v4/proofs/` and `research/model_v4/rederive/` for the two-pass evidence chains; `research/model_v4/HANDOFF_sign.md` and `quality_reports/fixes/t1_o1_rerun_check.json` for the O-1 numbers; `quality_reports/fixes/t2_*_check.json` for the check inventory; `numerical_v4/smoke_output.txt` for the smoke facts.